



Overestimating vs underestimating

Task A: Estimating with multiplication and addition

1) Identify if the following gives an **overestimate**, **underestimate** or if it is hard to determine.

- | | |
|--|--------------------------|
| a) $568.5445 + 453.56 \approx 570 + 500$ | Overestimate |
| b) $45.215 \times 62.123 \approx 50 \times 60$ | <i>Hard to determine</i> |
| c) $5321 \times 4236 \approx 5000 \times 4000$ | Underestimate |
| d) $2455 + 2362 \times 5466 \approx 2500 + 2400 \times 5500$ | Overestimate |
| e) $0.565 + 0.461 \times 0.31 \approx 0.6 + 0.5 \times 0.2$ | <i>Hard to determine</i> |
| f) $953 + 56.56 + 0.665 \times 0.6 \approx 1000 + 60 + 0.5 \times 0.5$ | Overestimate |
| g) $56.2^2 + 54.3^2 \approx 60^2 + 50^2$ | <i>Hard to determine</i> |

2) Here is a calculation

$$8\,467 \times 13.92 = 117\,860.64$$

a) Write a multiplication that gives a product close to, but a little more than, 117 860.64

Any values where the multiplier and/or multiplicand have been shifted upwards slightly. E.g. 8500×15

b) Write a multiplication that gives a product close to, but a little less than, 117 860.64

Any values where the multiplier and/or multiplicand have been shifted downwards slightly. E.g. 8400×10

Task B: Estimating with division and subtraction

1a) Without calculating,

$$15.99 \div 4$$

is this a little more than 4, a little less than 4 or exactly 4? *A little less than 4.*

b) Without calculating,

$$16 \div 3.99$$

is this a little more than 4, a little less than 4 or exactly 4? *A little more than 4.*



2) Identify if the following gives an overestimate, underestimate or if it is hard to determine.

a) $56.2 - 24.3 \approx 60 - 20$ *Overestimate*

b) $\frac{865}{123} \approx \frac{900}{100}$ *Overestimate*

c) $788.621 - 132.2 \approx 800 - 130$ *Overestimate*

d) $\frac{56+19}{0.36} \approx \frac{60+20}{0.2}$ *Overestimate*

e) $\frac{545+164}{14.5} \approx \frac{550+160}{15}$ *Underestimate*

f) $\frac{256}{573+546} \approx \frac{250}{575+550}$ *Underestimate*

g) $\frac{58-52}{0.59} \approx \frac{60-50}{0.5}$ *Overestimate*

h) $26.5^2 + 23.1^2 \approx 30^2 + 25^2$ *Overestimate*

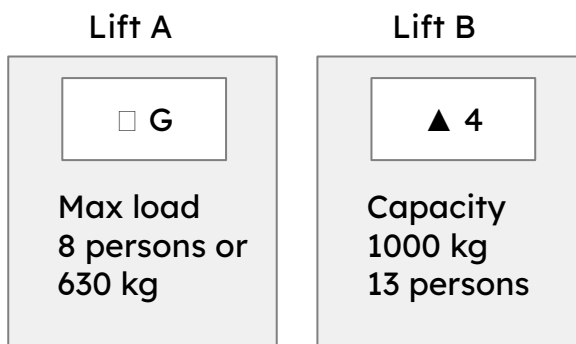
i) $\frac{256}{573-546} \approx \frac{300}{570-550}$ *Overestimate*

j) $\frac{26-5.6}{56-45} \approx \frac{30-6}{60-50}$ *Hard to tell*



Task C: Overestimating vs underestimating in context

The two signs on the left are from different lifts. Each lift can carry a different number of people.



a) Which lift assumes that people are heavier?

Lift A assumes people are heavier. A likely estimation might be $640 \div 8 = 80$ kg for Lift A and $1040 \div 13 = 80$ kg Lift B. As these give the same answer, the actual weight is closer to 80 for the first lift than it is for the second. The actual assumed weights are 78.75 kg and 76.92 kg (to 2 d.p.) respectively.

b) Which lift is easier to estimate? Why?

Answers can vary. Some may argue that the first lift is easiest, as 630 is close to 640, which is a multiple of 8, so $640 \div 80$ is a straightforward calculation. You may find the second lift easier may also use the strategy of finding a 'nearby' multiplication fact such as $960 \div 12$ or $1040 \div 13$.

c) Nine identical boxes are under the weight limit for one lift, but over it for the other lift. What do you know about the weight of the boxes?

The boxes must weigh between 70 kg and approximately 111 kg.

d) Five boxes weighing 23 kg each are put into each lift. Approximately how many people can safely get in the lifts with them?

Answers will depend on how to estimate or calculate with 23. You might take the total weight of the boxes to be 100 (rounding 23 to 20), 115 (using 23) or 125 (rounding 23 to 25). Approximately six people can get into the first lift with the boxes; approximately 11 people can get into the second lift with the boxes.

e) Explain why it is better to overestimate all the scenarios above.

*For estimating weight in a lift, it is better to **overestimate**. If we **underestimate**, then the lift is in danger of exceeding capacity and breaking.*